

Article

Ranking Influential Non-Content Factors on Scientific Papers' Citation Impact: A Multidomain Comparative Analysis

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Abstract: The influence of scientific papers is measured by their citations. Although predicting the papers' citation impact based on non-content factors has garnered extensive attention, the influence of such factors is rarely compared. In this article, we compare the influence of non-content factors on the citation counts of academic publications across three fields, i.e., math, computer science, and management. We consider different methods in this study, including three machine learning approaches, namely, XGBoost, Gradient Boosting Decision Tree, and Random Forest, along with statistical techniques such as linear regression and quantile analysis. Our findings reveal that no matter the field or analytical method applied, author prestige and the number of references consistently stand out as the most influential factors, while the breadth of categories covered by the paper has minimal impact. In mathematics, the first citation date and article length are almost equally important as author prestige, while the number of authors and the journal impact factor are crucial for computer science papers. In management, the number of collaborating countries is relatively influential with respect to the paper's citations. The results of the quantile regression indicate that at higher quantile levels, the impact of author prestige and the number of authors on the papers' citation impact are more pronounced across all three disciplines, while the journal impact factor and paper length have the greatest influence at low and medium quantile levels. Our findings indicate that the reliance of academic citations on author prestige and journal impact factors not only highlights the unequal distribution of resources within the current academic system but also further exacerbates citation inequality.

Keywords: citation impact; citation counts prediction; machine learning; quantile regression; scientific paper; author prestige



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1. Introduction

Scientific papers serve as a primary carrier of knowledge and a major propeller of innovation. The citation impact, also known as the citation count, is commonly defined as the total number of citations a paper receives [1] and is widely used to assess the scientific influence of publications.

While numerous studies have examined factors influencing citation impact, such as journals' impact factors [2], early-stage citations [3], readability [4], and the number of authors [5], two significant gaps remain in the current literature. First, there is a lack

of systematic comparison of these factors across academic fields and methodological approaches. Second, existing studies often focus on isolated disciplines, limiting our understanding of how these factors influence citation impact across different fields.

A noticeable gap lies in the limited comparison between different methods employed to predict citation impact factors, despite the considerable advancements in predictive accuracy achieved in recent years. Existing studies predominantly focus on enhancing the precision of citation count predictions using various methodologies, including quantile regression models [2,6,7], machine learning approaches [8–11], and transformer-based models [12,13]. For example, quantile regression has been widely utilized to explore citation distributions across diverse datasets, enabling insights into citation behaviors at specific percentiles. Machine learning methods, such as support vector machines and neural networks, have proven effective in capturing complex, non-linear relationships between predictors and outcomes, while transformer-based models leverage natural language processing (NLP) to analyze textual metadata and citation contexts, yielding state-of-the-art performance in identifying citation patterns. However, these studies typically evaluate the performance of a single method in isolation. Without systematic comparisons, the relative strengths and weaknesses of different approaches remain unclear, restricting the robustness of findings. For instance, while machine learning techniques demonstrate flexibility in handling large datasets [14], multi-feature models provide a holistic perspective by integrating metadata and journal-specific attributes [15], and regression models offer interpretable relationships between predictors and citation outcomes [7]. Systematic evaluation of diverse predictive frameworks across varied contexts is needed to ensure methodological robustness and generalizability.

Another noteworthy limitation is the insufficient cross-disciplinary comparison of influential factors affecting citation impact, particularly non-content factors. While prior studies have explored how factors such as author reputation, institutional prestige, and collaboration patterns influence citation counts, these analyses are often confined to single academic disciplines. For instance, investigations have focused on medicine [5,10], operations research [16], and economics [17], providing valuable but field-specific insights. Such discipline-bound analyses fail to address a critical question: whether the same non-content factors consistently influence different fields. For example, while institutional reputation may play a significant role in economics, collaboration intensity may have a greater impact in biomedical sciences. Without systematic cross-disciplinary comparisons, the generalizability of findings remains uncertain, and opportunities to identify field-specific or universal patterns are missed. This gap underscores the need for comparative studies spanning multiple disciplines to identify overarching trends and field-specific nuances. Such efforts would not only enhance the theoretical foundations of citation analysis but also inform strategies for improving research visibility and impact across academic fields.

To address these gaps, this study explores how non-content factors, such as author prestige, journal impact factors, and collaboration metrics, influence the citation impact of scientific papers across mathematics, computer science, and management. Gaining insight into these dynamics is essential for guiding research strategies, optimizing reference selection, and refining evaluation metrics within academia.

Notably, machine learning methods have demonstrated remarkable versatility in addressing domain-specific problems across diverse scientific fields. For instance, Peddakapu et al. employed an Adaptive Neuro-Fuzzy Inference System (ANFIS) to identify faults in high-voltage transmission lines, outperforming traditional methods [18]. In aerial image processing, Liu et al. highlighted the high accuracy of Convolutional Neural Networks (CNNs) for object detection and classification [19]. Similarly, Yang et al. utilized deep learning to model Alzheimer's disease progression, integrating multimodal data for clinical

applications [20]. In the field of nanofluid mechanics, Zhang et al. compared Multi-Layer Perceptron (MLP); Radial Basis Function (RBF); and Recursive Least Squares (RLS) fuzzy models to predict the properties of CuO/liquid paraffin mixtures, demonstrating the strengths of each approach [21]. These studies underscore the importance of method selection based on the specific requirements and data characteristics of each problem, further emphasizing the need for cross-disciplinary comparisons in citation impact research.

This study aims at comparing the factors influencing citation impact via a comprehensive investigation of academic papers across mathematics, computer science, and management. Methodologically, this comparison is based on three machine learning methods, including Extreme Gradient Boosting (XGBoost, version 1.7.3); Gradient Boosting Decision Tree (GBDT, using sklearn version 1.3.2); and Random Forest (RF, using sklearn version 1.3.2); followed by multiple linear regression and quantile regression. Machine learning (ML) methodologies were chosen for this study due to their ability to handle complex, non-linear relationships, to adapt to high-dimensional data structures, and to mitigate overfitting through robust feature selection mechanisms [22]. While traditional statistical approaches, such as linear regression and quantile regression, provide valuable insights into the mean and conditional distribution of citation impact, ML techniques, including ensemble learning models like XGBoost and Random Forest, excel at uncovering hidden patterns in data without requiring explicit assumptions about the functional form of relationships. Additionally, the flexibility of ML models allows for the integration of diverse features, such as author prestige, journal impact factor, and collaboration metrics, which are crucial for understanding the multifaceted drivers of citation impact. All these methods have been applied to citation prediction and have proven effective in improving prediction accuracy [8,11]. Combining these techniques, our method addresses certain limitations in existing studies and facilitates a more detailed analysis of the factors influencing citation impact. The comparative evaluation of ML methodologies, alongside traditional regression approaches, provides robust insights into their applicability for predictive and interpretative purposes in bibliometric research, addressing the need for advanced tools in navigating the complexities of citation systems across disciplines.

We aggregate a unique and extensive dataset from prominent journals in each of the three included fields, including those cataloged by the American Mathematical Society (AMS); the Association for Computing Machinery (ACM); and the 24 journals curated by the University of Texas at Dallas (UTD24). The data span from the inception of these journals to 20 June 2023. Structured data include metrics such as citation counts, author prestige, the number of authors, the number of collaborating countries, journal impact factors, research fields, and the first citation date. We preprocess the original dataset, followed by normalization and encoding transformations to standardize variables across different ranges. Linear regression and quantile regression further enhance the robustness of the results. While linear regression models provide insights into the mean effects of variables, quantile regression allows us to flexibly examine variable relationships at different quantiles, better capturing the diversity of influencing factors across varying citation levels.

The contributions of this paper are as follows. First, this study conducts a comprehensive comparison from both methodological and disciplinary perspectives and effectively reveals the underlying inequalities in citation practices. By applying multiple linear regression, quantile regression, and three machine learning techniques, namely, XGBoost, GBDT, and Random Forest, and analyzing three fields, namely, mathematics, computer science, and management, this study highlights disparities across methods and disciplines and, through data and empirical analysis, uncovers key mechanisms that drive citation inequalities. Second, based on extensive literature analysis and research data, this paper

provides evidence and theoretical insights demonstrating how author prestige exacerbates citation inequality. The study finds that prestigious authors leverage their academic influence and extensive collaboration networks to secure journal acceptance and resource support, which amplifies their visibility and citation counts through a positive feedback mechanism. Preferences for prestigious authors in citation and editorial decision-making further magnify citation inequality. These logical pathways help explain the deeper causes behind the phenomenon and offer clear directions for future research. Quantile regression reveals differences in the impact of various factors across citation levels and provides deeper insights into the diversity within citation distributions. These contributions emphasize the need for a more equitable academic evaluation system that moves beyond reliance on identity-based metrics and promotes fairer practices in citation analysis and scholarly recognition.

The remainder of the paper is organized as follows: Section 2 elaborates on the research methods; Section 3 explains the data collection process; Section 4 presents the empirical results; Section 5 presents the discussion; and Section 6 is the conclusion.

2. Methods

2.1. Machine Learning

This study employs three tree-based ensemble learning methods, namely, Extreme Gradient Boosting, Gradient Boosting Decision Tree, and Random Forest, to predict the papers' citation impact in mathematics, computer science (CS); and management (Mgmt). These models were selected based on their ability to handle diverse data types, robustness to outliers and non-linearity, and proven effectiveness in citation prediction [23]. Tree-based models excel in processing structured data and categorical variables, making them well-suited for datasets involving a mix of continuous and discrete features, such as the citation impact dataset used in this study. Additionally, these models naturally provide insights into feature importance, allowing us to rank factors like author prestige, paper length, and journal impact factor. Random Forest uses random sampling of both data points and features, while XGBoost and GBDT include regularization terms, making them highly effective in reducing overfitting in predictive models. These methods have been extensively validated in prior studies on citation prediction, demonstrating their ability to improve model performance through feature selection and ensemble strategies.

GBDT establishes a robust predictive model by integrating multiple Decision Trees. Each tree learns from the residuals of the predictions made by the previous tree, addressing errors from earlier models. This process allows the model to improve its accuracy step by step [24]. GBDT considers the cumulative predictions of all previous trees when constructing each new tree, a strategy that helps gradually reduce errors during training. XGBoost builds multiple Decision Trees and combines them to enhance prediction accuracy. At each step, XGBoost uses a gradient boosting framework to reduce model error and includes regularization terms in the objective function to control model complexity, effectively preventing overfitting [25]. Random Forest operates by constructing multiple Decision Trees and integrating their predictions. During training, each tree randomly selects samples and features from the original dataset, imparting robustness to the model and effectively resisting overfitting through this randomness [26]. All the machine learning methods have been applied to predict the impact of articles [8,14,27]. The principal flowchart of the three machine learning algorithms (GBDT, XGBoost, and RF) used for regression analysis is presented in Figure 1.

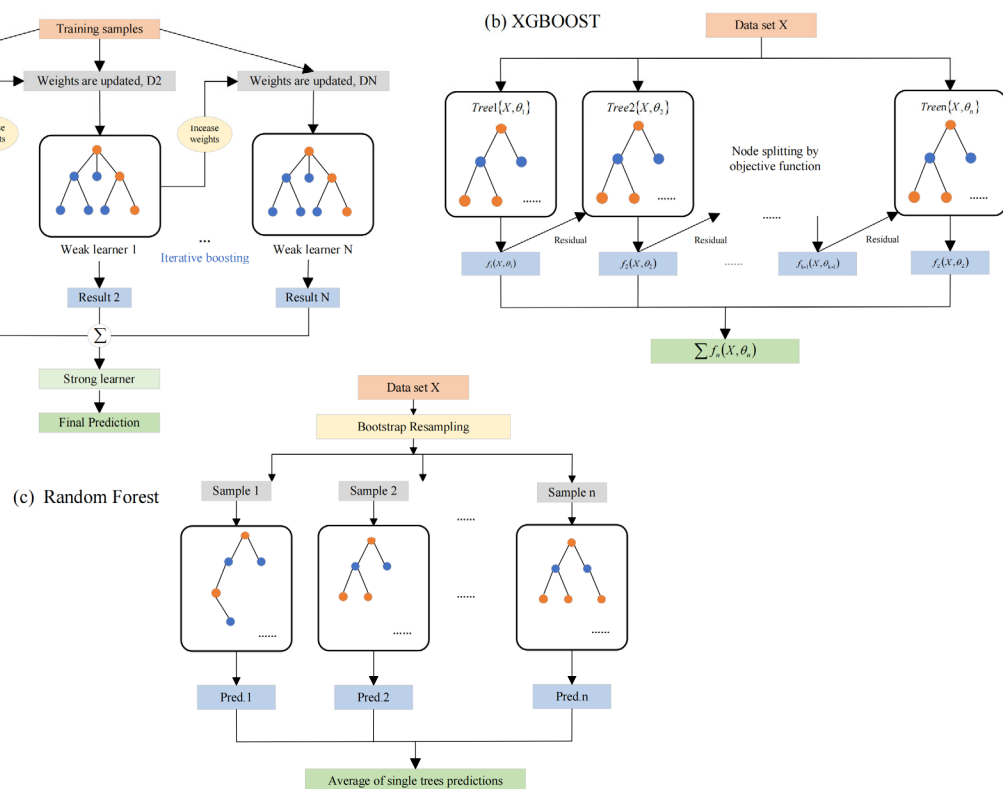


Figure 1. (a) Flowchart of the GBDT algorithm. (b) Flowchart of the XGBoost algorithm. (c) Flowchart of the RF algorithm.

2.2. Linear Regression

Linear regression was selected for its statistical interpretability and role as a baseline model in this study. As a classical approach, linear regression identifies the mean effect of independent variables on the dependent variable, providing straightforward and interpretable coefficients. This facilitates a clear understanding of how factors such as author prestige and journal impact factor influence citation counts. Additionally, linear regression serves as a benchmark for comparing the predictive power of more complex methods, such as machine learning models. This comparison helps determine whether advanced techniques significantly improve prediction accuracy. Scholars have convincingly demonstrated the effectiveness of multiple linear regression models in predicting paper citations [8,17,28], making it a reliable choice for establishing a solid statistical foundation for further analysis and discussion.

We employ a multiple linear regression model to predict citation impact. This approach provides a solid statistical foundation for further analysis and discussion. The form of the multiple linear regression model is as follows:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_n X_n \quad (1)$$

where Y represents the dependent variable, and $X_1, X_2, \dots, X_n$, represent the independent variables.

2.3. Quantile Analysis

Quantile regression was selected for its ability to provide a nuanced understanding of citation distributions, particularly in the context of highly skewed and zero-inflated data [6]. Unlike traditional linear regression, which focuses solely on the conditional mean, quantile regression offers a flexible approach to analyze how independent variables influence

different parts of the citation distribution. This is especially valuable for identifying factors that drive high-citation papers (upper quantiles) and revealing the characteristics of low-citation papers (lower quantiles); such as journal impact factor or first citation date. Citation counts are typically right-skewed, with a few papers receiving disproportionately high citations, making quantile regression more robust and suitable for such distributions. Additionally, quantile regression allows us to capture discipline-specific variations in citation patterns, such as the impact of the number of authors in management or paper length in mathematics, across different citation levels. By combining quantile regression with traditional statistical techniques and machine learning models, this study ensures a comprehensive and robust analysis, addressing multiple research questions effectively.

The Matthew Effect posits that when there are sharp and noticeable differences in prestige within the academic hierarchy, a scientific contribution will receive disproportionate peer recognition [29]. Papers or authors with higher prestige are more likely to be cited. Due to the presence of the Matthew Effect, the distribution of citation counts tends to be highly right-skewed and zero-inflated, leading to poor performance of traditional prediction methods [30]. For such distributions, the mean is heavily influenced by outliers and may be a misleading measure of central tendency [2]. We are particularly interested in the factors influencing higher quantiles, as high-citation studies are often of greater concern. Quantile regression provides a non-parametric method to analyze the relationship between different quantiles of the dependent variable distribution and the independent variables. The form of the quantile regression model is as follows:

$$Q_Y(\tau) = \beta_0(\tau) + \beta_1(\tau)X_1 + \cdots + \beta_n(\tau)X_n \quad (2)$$

where $Q_Y(\tau)$ represents the τ -th quantile of the dependent variable Y , and τ is a value between 0 and 1.

3. Data

3.1. Variable Definition

In selecting variables to predict the papers' citation impact, we consider the following factors. First, we choose author prestige, defined as the highest average citation count of articles published by all co-authors, because authors' academic prestige is often closely linked to the impact of their papers. Second, we consider the number of co-authors and the number of collaborating countries, which reflect collaboration and internationalization, potentially influencing citation counts significantly. We also include factors contributing to the paper's depth, richness, and scholarly impact, such as the breadth of categories covered and the references cited. Additionally, paper length may indicate research depth and detail, potentially impacting citation rates. Finally, we incorporate the impact factor of the publishing journal and the first citation date, as the journal impact factor reflects academic prestige and influence, while the first citation date reflects the speed and persistence of a paper's impact in academia. These variables are selected based on understanding their potential effects on citation impact across different dimensions and hypotheses. All variables are as shown in Table 1.

Table 1. Definitions of the selected variables.

Variable	Definition	Prior Literature
Prestige	The highest average citation count of articles published by the most renowned co-author. It is calculated by computing the average citation count of all articles published by each co-author and identifying the highest value among these averages.	Gao et al. [30]
N_auths	The total number of co-authors involved in a paper.	Li et al. [10]
N_cntrs	The number of different countries represented by the researchers involved in the paper.	Li et al. [10]
N_catg	The diversity of research topics or fields covered by the paper, calculated by counting the number of distinct classification labels associated with the paper.	Annalingam et al. [31]
N_kwds	The total number of keywords.	Chakraborty et al. [32]
N_refs	The number of references cited in the paper.	Hu et al. [9], Ruan et al. [11]
L_pages	The actual length of the paper.	Hu et al. [9], Ruan et al. [11]
IF	Impact factor of the journal in which the paper is published.	Bornmann and Williams [33]
First	Time elapsed since the paper's publication until its first formal citation by another work.	Bai et al. [15]

3.2. Data Collection and Preprocessing

Web of Science is widely recognized for its extensive coverage and authoritative citation data across various scientific disciplines, making it a reliable source for scholarly research and analysis [34]. The dataset used for calculation and validation in this study is drawn from the Science Citation Index Expanded, which is accessible through the Web of Science platform. The research direction is a set of classification attributes used by papers in all databases under the Web of Science, designed to identify, retrieve, and analyze the literature on the same topics across multiple databases. This study involves three fields: mathematics, computer science, and management.

In the field of mathematics, the literature comes from all journals under the American Mathematical Society (AMS); an authoritative and representative body in mathematical research that publishes high-quality journals covering a broad range of mathematical topics [35]. The AMS is known for its rigorous peer-review process and significant contributions to the advancement of mathematics. In the field of computer science, the literature is sourced from all journals under the Association for Computing Machinery (ACM). ACM is the world's largest educational and scientific computing society, known for its leading publications and conferences that foster innovation and excellence in computing and information technology [36]. For management, the literature comes from the UTD24, a set of journals recognized for their high impact on management research. The UTD24 journal list is widely used as a benchmark for evaluating research quality and impact in the management discipline [37]. The publication dates of all relevant literature range from the inception of each journal to 20 June 2023. Table A1 below summarizes the journals included in this study.

According to the definition of the research question, t was set at 1 year, 3 years, and 10 years, respectively, and the actual citation frequency during the t period after the publication of the papers was compared.

For structured data, the main elements we process include metrics such as citation counts, the number of authors, the number of collaborating countries, the impact factor of the publishing journal, the breadth of categories covered by the paper, and the first citation date. Initially, we perform data cleaning, which involves handling missing values, outliers, and duplicates to ensure the completeness and usability of the data. Subsequently, we perform normalization and one-hot encoding transformations for these continuous

and categorical indicators. This is undertaken to eliminate the impact of differences in numerical ranges between various indicators on model training and to make them suitable for input into the model.

3.3. Descriptive Analysis

Table 2 shows how the distribution of citation counts and the non-content factors influencing citations vary across mathematical, computer science, and management journals. In the field of mathematics, the distribution of citation counts shows significant variation, with a small number of highly cited papers causing the overall distribution to skew, indicating a high degree of imbalance. For instance, the mean citation count is 12.92, but the maximum reaches as high as 2441, highlighting this imbalance. The pronounced skewness and kurtosis suggest that the high citation counts of certain papers may be closely related to factors such as high author prestige, a large number of references, and longer paper length. The distribution of these factors also exhibits considerable variability, indicating that differences in non-content factors across papers may have a significant impact on citation outcomes.

Table 2. Descriptive statistics of non-content factors influencing citation impact.

Field	Statistics	Citation Counts	Prestige	N_auths	N_cntrs	N_catg	N_kwds	N_refs	L_pages	IF	First
Math (33,641)	Mean	12.92	15.77	1.78	1.18	2.08	2.16	20.36	18.79	1.32	4.52
	Std	33.11	31.09	0.88	0.83	0.92	2.51	17.28	20.52	0.52	4.11
	Min	0	0	0	0	1	0	0	0	0.246	0
	Median	5	8	2	1	2	1	16	13	1	4
	Max	2441	1449	13	6	4	10	405	346	4.8	12
	Kurt	1123.4	372.55	3.26	1.5	−1.78	1.48	50.09	34.59	2.03	−1.28
	Skew	21.8	12.95	1.29	0.85	−0.15	1.41	4.68	4.72	1.27	0.37
CS (40,802)	Mean	33.37	50.72	3.68	1.47	3.14	1.55	42.08	20.5	4.17	6.6
	Std	224.57	233.14	2.13	0.88	1.45	2.28	33.17	12.11	4.59	3.48
	Min	0	0	0	0	1	0	0	0	0.1	0
	Median	8	23.25	3	1	2	1	36	18	2.9	7
	Max	26,946	26,032	49	13	8	10	892	169	21.6	12
	Kurt	8285.09	6134.83	26.82	7.74	1.65	3.86	26.33	3.31	4.14	−1.05
	Skew	78.96	66.42	3.05	1.73	1.28	1.99	3.47	1.2	2.08	−0.19
Mgmt (41,110)	Mean	113.93	143.1	2.55	1.62	2.14	6.8	57	21.48	9.14	6.55
	Std	291.84	250.98	1.1	0.94	0.74	3.41	34.16	9.56	3.49	3.55
	Min	0	0	1	0	0	0	0	0	2.5	0
	Median	40	78.44	2	1	2	8	50	19	8.4	6
	Max	15,903	14,394	49	29	4	10	536	115	17	12
	Kurt	533.82	454.92	146.19	34.73	−0.29	−1.08	9.03	2.06	−0.77	−1.24
	Skew	15.83	13.79	4.91	2.35	0.21	−0.59	1.88	1.2	0.19	0.02

In contrast, the distribution of citation counts in computer science spans a broader range, with some papers receiving exceptionally high citation counts, leading to a highly skewed distribution. The extreme variability in citation distribution within this field is likely associated with non-content factors such as author prestige, number of authors, and number of references. These factors appear to play a critical role in influencing the citation impact of papers in computer science, particularly in cases involving large research teams and extensive collaborations.

The field of management demonstrates a more balanced citation pattern. While papers generally receive a higher average number of citations, the lower skewness in distribution indicates a more uniform impact across papers. Non-content factors such as the number of keywords and paper length may reflect the field's emphasis on comprehensiveness and

depth in research, with the citation impact being more evenly distributed across various research topics.

4. Result

4.1. Outcomes of Machine Learning

Table 3 presents a comparative analysis of the primary performance metrics for the XGB, GBDT, and RF machine learning models in forecasting citation impact across mathematics, computer science, and management. Across all disciplines, the R-square values for these models consistently approximate 0.7, demonstrating comparable and substantial explanatory capacity. This indicates that each model accounts for approximately 70% of the data variance, effectively capturing the primary trends. While these R^2 values may appear modest, they align with the inherent complexity of citation dynamics, reflecting the multi-faceted nature of academic influence. The consistent performance of GBDT, XGBoost, and Random Forest across disciplines highlights the robustness of our approach, capturing the primary trends in citation impact. For practical applications, the R^2 values obtained in this study provide meaningful and actionable insights, even with some unexplained variability.

Table 3. Performance metrics of machine learning methods for predicting citation impact.

Field	Method	R-Square	MSE	MAE
Math	XGB	0.740	284.638	7.007
	GBDT	0.781	240.255	6.577
	RF	0.765	257.706	6.530
CS	XGB	0.800	5033.764	23.864
	GBDT	0.836	3236.544	21.423
	RF	0.834	8392.746	29.072
Mgmt	XGB	0.685	26,797.432	67.814
	GBDT	0.725	23,430.657	65.257
	RF	0.721	23,771.784	64.996

Overall, the GBDT model outperforms both the XGB and RF models across all fields, demonstrating stronger explanatory power and predictive accuracy. The differences in model performance across disciplines reflect the varying strengths and limitations of each model in capturing citation impact.

Table 4 presents the contribution of coefficients for predicting citation impact using the XGB, GBDT, and RF machine learning models across the disciplines of mathematics, management, and computer science. In all three fields, regardless of the model used, prestige consistently emerges as the most influential factor for citation counts, indicating that co-authors' prestige plays a critical role in driving citation frequency. This finding underscores a well-documented phenomenon in academia whereby high-prestige authors or institutions typically attract more citations. However, the relative weight of each factor varies across disciplines, likely influenced by the distinct academic norms and research cultures that shape citation practices and priorities within each field.

This "prestige effect" reflects the broader academic trend whereby established researchers, often associated with leading institutions or high-impact work, enjoy higher visibility and influence, which naturally elevates their citation counts. The consistency of these findings across multiple machine-learning models underscores their robustness and presents opportunities for further research on how these factors interact within specific disciplinary frameworks, potentially revealing strategies for maximizing citation impact tailored to each field's unique conventions.

Table 4. Coefficient values of critical factors influencing citation counts.

Feature	Math			CS			Mgmt		
	XGB	GBDT	RF	XGB	GBDT	RF	XGB	GBDT	RF
Prestige	0.5566	0.7471	0.7475	0.571	0.6695	0.8553	0.4552	0.8676	0.6625
N_auths	0.0297	0.0263	0.0207	0.0433	0.0307	0.019	0.0563	0.0134	0.0326
N_cntrs	0.0673	0.0247	0.0204	0.0791	0.017	0.0083	0.1236	0.0049	0.0241
N_catg	0.107	0.0079	0.0107	0.0555	0.0076	0.0061	0.0083	0.0061	0.0124
N_kwds	0.0448	0.0252	0.0255	0.0947	0.0259	0.0109	0.0568	0.009	0.0264
N_refs	0.0559	0.0612	0.0675	0.0319	0.0728	0.0407	0.0569	0.0598	0.0771
L_pages	0.0268	0.0482	0.0544	0.0325	0.1022	0.0287	0.1187	0.0197	0.0815
IF	0.0386	0.0159	0.0148	0.092	0.0377	0.0137	0.0654	0.0096	0.0403
First	0.0734	0.044	0.0386	0.0052	0.0367	0.0175	0.067	0.01	0.0431

4.2. Outcomes of Regression Analysis

The results obtained from the multiple linear regression model in Table 5 indicate that prestige and journal impact factor have a positive and statistically significant effect on citation counts across the disciplines of mathematics, management, and computer science. This finding suggests that papers authored by renowned researchers or published in high-impact journals tend to receive more citations, reflecting the widely recognized “Matthew Effect” in academia, where author prestige plays a crucial role in scientific evaluation.

Table 5. Regression results.

Feature	Math	CS	Mgmt
Prestige	0.4961 *** (227.568)	0.795 *** (330.088)	0.7972 *** (231.861)
N_auths	−0.0096 *** (−12.227)	−0.0074 *** (−13.936)	−0.0464 *** (−16.405)
N_cntrs	−0.0015 *** (−3.794)	−0.0007 * (−1.916)	−0.0084 *** (−4.167)
N_catg	−0.0006 *** (−3.422)	−0.0002 * (−1.726)	0.0011 *** (3.056)
N_kwds	−0.0012 *** (−4.974)	0.0006 *** (5.117)	−0.0005 ** (−2.136)
N_refs	0.0237 *** (14.937)	−0.0031 *** (−3.605)	0.0083 *** (6.941)
L_pages	−0.0037 *** (−3.609)	0.002 *** (5.376)	0.0005 (0.67)
IF	0.0011 ** (2.309)	0.0007 *** (6.066)	0.0011 *** (3.859)
First	−0.0004 *** (−2.626)	−0.0002 ** (−2.459)	−0.0003 (−1.455)
Constant	0.0009 *** (4.721)	0.0002 ** (2.136)	−0.0004 (−1.325)
Observations	33,641	40,802	41,110
R-squared	0.625	0.730	0.588

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

In contrast, the coefficients for the number of authors and the number of collaborating countries are negative and statistically significant across all three fields, indicating that papers with more co-authors or more diverse international collaborations tend to have lower citation counts. This may be due to the perception that papers with numerous authors or cross-country collaborations could lack a cohesive perspective or focus, potentially diluting the paper’s impact in terms of perceived originality or authority. In academic fields where

individual contribution and focused expertise are highly valued, such as management science, this trend might be more pronounced, as single-authored or small-team research is often seen as more targeted and authoritative. While other variables in the model also show statistical significance, their influence on citation counts varies by discipline. This variation highlights how different academic fields prioritize unique factors in citation practices.

The results of the quantile regression analysis in Table 6 reveal that the positive impact of author prestige and the negative impact of author count on citation impact become more pronounced at higher quantile levels across mathematics, management, and computer science. This suggests that for highly cited papers, author prestige is a key factor in amplifying citation impact. High prestige authors are more likely to gain recognition and resource support as their research is often perceived as more credible or of higher quality, making it more likely to be cited to enhance a citing paper's persuasiveness. This accumulation of prestige grants these scholars an advantage in new publications, establishing a positive feedback loop of cumulative advantage. Conversely, emerging or less well-known researchers find it challenging to receive comparable attention, leading to a concentration of resources and recognition among a small group of scholars and, consequently, exacerbating inequalities in academia. Papers with multiple authors may not receive the same level of recognition for each contributor, especially in high-citation contexts, where individual contributions might be less identifiable. Larger teams of authors may detract from individual recognition, potentially diluting perceived authorship influence.

Table 6. Quantile regression results in mathematics, computer science, and management science.

Domain	Quantile	Prestige	N_auths	N_cntrs	N_catg	N_kwds	N_refs	L_pages	IF	First	Constant
Math	Q10	0.083 *** (168.780)	−0.061 *** (−3.467)	−0.041 ** (−2.220)	−0.031 * (−1.823)	0.011 (1.605)	0.000 (0.380)	−0.004 *** (−4.838)	0.161 *** (5.348)	−0.009 *** (−2.947)	0.029 (0.399)
	Q30	0.410 *** (712.662)	−0.302 *** (−13.184)	−0.204 *** (−8.274)	−0.137 *** (−6.071)	0.024 ** (2.531)	0.009 *** (5.850)	−0.008 *** (−7.282)	0.622 *** (15.386)	−0.031 *** (−7.429)	0.511 *** (5.522)
	Q50	1.000 *** (4527.902)	−0.333 *** (−37.684)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.333 *** (9.667)
	Q70	1.013 *** (6290.339)	−0.053 *** (−9.172)	−0.017 *** (−2.769)	−0.007 (−1.253)	−0.002 (−0.823)	0.002 *** (4.431)	−0.001 *** (−1.863)	0.005 (0.449)	−0.001 (−0.691)	0.079 *** (3.610)
	Q90	1.355 *** (1140.784)	−0.888 *** (−16.716)	−0.471 *** (−8.828)	−0.202 *** (−4.448)	−0.082 *** (−4.193)	0.055 *** (16.308)	−0.008 *** (−3.407)	0.519 *** (6.570)	−0.013 (−1.504)	2.125 *** (11.945)
	Q10	0.013 *** (127.911)	−0.025 *** (−2.761)	−0.017 (−0.752)	−0.030 ** (−2.430)	0.020 ** (1.972)	−0.002 * (−1.748)	0.006 *** (3.054)	0.011 ** (2.127)	0.000 (−0.002)	0.025 (0.339)
CS	Q30	0.161 *** (1216.765)	−0.310 *** (−18.672)	−0.200 *** (−4.963)	−0.184 *** (−7.947)	0.166 *** (9.213)	−0.010 *** (−6.817)	0.051 *** (15.201)	0.164 *** (19.428)	−0.016 (−1.616)	0.900 *** (6.884)
	Q50	0.581 *** (2638.976)	−1.001 *** (−38.587)	−0.719 *** (−11.509)	−0.118 *** (−3.210)	0.284 *** (10.256)	−0.007 *** (−3.190)	0.086 *** (16.669)	0.150 *** (12.013)	−0.061 *** (−4.122)	1.778 *** (8.888)
	Q70	1.035 *** (6453.634)	−0.774 *** (−44.956)	−0.518 *** (−12.544)	−0.049 ** (−2.014)	0.094 *** (5.158)	−0.004 ** (−2.476)	0.037 *** (11.084)	0.018 ** (2.252)	−0.034 *** (−3.463)	1.556 *** (11.871)
	Q90	1.167 *** (9465.189)	−0.811 *** (−25.434)	−0.549 *** (−7.744)	−0.067 * (−1.660)	−0.063 ** (−2.194)	0.003 (1.063)	0.050 *** (8.426)	0.106 *** (8.166)	−0.010 (−0.630)	3.075 *** (14.063)
	Q10	0.108 *** (258.834)	−1.663 *** (−18.752)	−1.256 *** (−11.540)	0.138 (0.935)	0.339 *** (9.851)	−0.037 *** (−11.482)	−0.119 *** (−10.292)	0.380 *** (12.549)	0.034 (1.286)	3.199 *** (6.612)
	Q30	0.367 *** (598.557)	−4.212 *** (−28.248)	−2.581 *** (−14.255)	0.923 *** (3.890)	0.336 *** (5.798)	−0.026 *** (−4.551)	−0.238 *** (−12.536)	0.788 *** (15.919)	−0.004 (−0.093)	7.990 *** (10.116)
Mgmt	Q50	0.781 *** (946.989)	−6.431 *** (−31.602)	−3.448 *** (−14.376)	2.340 *** (7.712)	0.007 (0.096)	−0.002 (−0.307)	−0.214 *** (−8.699)	0.482 *** (7.411)	−0.059 (−1.047)	10.679 *** (10.381)
	Q70	1.030 *** (1880.218)	−3.873 *** (−29.622)	−1.408 *** (−9.599)	1.032 *** (5.709)	0.023 (0.493)	0.004 (0.922)	−0.082 *** (−5.448)	0.131 *** (3.287)	−0.059 * (−1.730)	7.365 *** (11.727)
	Q90	1.545 *** (1109.625)	−8.083 *** (−17.287)	−1.619 *** (−3.756)	3.176 *** (6.230)	−0.335 *** (−2.602)	0.101 *** (7.541)	−0.001 (−0.022)	−0.015 (−0.140)	−0.241 ** (−2.498)	14.348 *** (7.655)

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

In contrast, journal impact factor and paper length have the greatest influence at low and medium quantile levels, indicating that for papers with fewer citations, publishing in a high-impact journal and providing substantial content are critical to gaining visibility. This pattern suggests that, especially for less-cited papers, journal reputation and paper length can serve as indicators of quality, helping to attract attention in an otherwise competitive academic environment. For early-career researchers or those with lesser-known profiles, publishing in reputable journals or producing comprehensive studies could be effective strategies for increasing citation potential. This also implies that the emphasis on journal

impact factor may prevent high-quality research published in lower-IF journals from receiving deserved attention, thereby further exacerbating citation inequalities.

The quantile regression analysis in Table 6 reveals several distinct trends in how various factors influence citation counts in the field of mathematics. The number of co-authors demonstrates a significant negative impact on citation counts across all quantiles, with this effect becoming stronger at higher quantile levels. This suggests that in mathematics, papers authored by individuals or small teams are more likely to receive citations, potentially because single-author or small-team works are often perceived as having more focused contributions or clearer intellectual ownership. In this field, individual expertise and originality are highly valued, which might explain why papers with fewer authors tend to attract more citations, especially at high citation levels.

Similarly, the number of collaborating countries significantly negatively impacts citation counts at all quantiles, particularly for highly cited papers. This may reflect a preference within the mathematics community for locally grounded research or less dependent on extensive international collaboration, which could sometimes dilute the perceived focus of the work. Likewise, the breadth of categories covered by the paper negatively affects citation counts at the Q30 and Q90 quantiles, indicating that papers with a narrower, more concentrated research theme are more likely to attract citations. This aligns with the mathematical field's emphasis on depth and specificity, where targeted research is often more appreciated and cited.

At the Q30 quantile, the number of keywords positively impacts citation counts, highlighting the importance of carefully selecting keywords to increase visibility for papers with lower citation impact. For less-cited papers, effective keyword usage may be critical in reaching a broader audience, ensuring that the paper appears in relevant search results and attracts attention. Furthermore, the number of references cited in the paper positively affects citation counts at both the Q30 and Q90 quantiles. This suggests that reference quantity is crucial for papers at both lower and higher citation levels, likely because a well-cited work is perceived as more thorough and credible. This trend is consistent with the expectation that a comprehensive literature review enhances the paper's credibility, making it a reliable resource that other researchers are more likely to cite.

The journal impact factor also shows a positive and significant influence across different quantiles, with the highest coefficient at the Q30 quantile, indicating that for papers with low-to-moderate citation counts, publishing in a high-impact journal is especially beneficial for visibility. This reflects a broader trend in academia whereby journal reputation can be a proxy for quality, especially for papers with fewer citations. Lastly, the first citation date has a significant negative impact on citation counts for papers with lower citation impact, indicating that papers that receive earlier citations are more likely to accumulate citations over time. However, this effect diminishes for highly cited papers, suggesting that once a paper gains substantial citations, the timing of initial citations becomes less relevant.

In the field of computer science, the quantile regression analysis in Table 6 reveals several unique trends that highlight the discipline's emphasis on innovation, technical precision, and the influence of prominent research teams. At higher quantile levels, the coefficient for prestige increases significantly, indicating that co-authors' prestige becomes particularly influential in highly cited papers. This reflects the common practice in computer science of citing works by well-known researchers or prominent teams, as their established prestige often contributes to the perceived credibility and visibility of the research. The negative impact of the number of authors on citation counts across all quantiles, with a stronger effect at higher quantiles, suggests that papers by single individuals or small teams are more likely to receive citations. Smaller teams may be better positioned to deliver focused contributions in a field where originality and technical depth are highly

valued. Papers from smaller teams are often perceived as more cohesive and directly attributable to individual expertise, which could enhance their appeal and citation impact, especially for highly cited studies. The number of collaborating countries also significantly negatively affects citation counts, particularly from the Q50 to Q90 quantiles. This may indicate that, in computer science, papers involving extensive international collaboration are sometimes seen as less cohesive in research focus. Computer science often values problem-solving within specific technical domains, and diverse international input may dilute the specialization required for focused technical advancements, thereby reducing citation potential. The breadth of categories covered by the paper also shows a significant negative impact on citation counts across all quantiles, suggesting that computer science papers with a more specialized focus are more likely to be cited. This aligns with the field's emphasis on depth and problem-specific solutions, where studies targeting particular technical challenges are more likely to gain traction among scholars. The number of keywords in the paper positively impact citation counts from the Q10 to Q70 quantiles, with the strongest effects in the middle quantiles. This result underscores the role of strategic keyword usage in enhancing a paper's visibility in computer science, where discoverability in search engines and databases can significantly influence a paper's reach. For highly cited papers, however, the number of keywords appears less influential, possibly because the established prestige of authors and journals becomes the primary factor driving citation impact. In contrast, paper length positively affects citation counts across all quantiles, with the effect being stronger at higher quantiles. This suggests that longer papers, which typically provide more comprehensive explanations or detailed technical insights, are more likely to attract citations in computer science. This reflects the academic preference for substantial content in high-impact research within this field. Lastly, the journal impact factor has varying effects across quantiles, with a stronger positive influence at the Q30 and Q50 quantiles. For moderately cited papers, publishing in high-impact journals plays an important role in increasing visibility. This pattern aligns with the competitive nature of computer science research, where publication in reputable journals is often a critical factor for gaining recognition, particularly for studies aiming to establish a strong academic presence.

In the field of management, the quantile regression results in Table 6 reveal a distinct citation pattern that emphasizes interdisciplinary reach and the strategic role of author prestige. Prestige significantly influences all quantiles, with an increasing effect at higher quantiles. This suggests that in management, high-citation papers particularly benefit from the visibility and credibility of reputable co-authors, aligning with the field's appreciation for established expertise in navigating complex organizational and market dynamics. Unlike in mathematics and computer science, the breadth of categories covered positively impacts citation counts across all quantiles, with a stronger effect at higher quantiles. This indicates that management research benefits from interdisciplinary or wide-ranging studies that address diverse topics, resonating with the field's broader approach to practical application and cross-disciplinary insights. Author count significantly negatively affects citations at all quantiles, especially at higher quantiles, reflecting a preference for research perceived as more streamlined and directly attributable. Similarly, more collaborating countries tend to reduce citation counts, particularly at the Q50 quantile. This may imply that highly collaborative international projects are less prominent in this field's citation landscape, possibly due to a focus on region-specific insights or streamlined management practices. The number of keywords significantly enhances citation counts from Q10 to Q30, highlighting the importance of effective keyword strategy for visibility among lower-cited management papers. Optimizing keyword relevance for papers in this citation range is particularly useful in attracting a focused audience in management databases and repositories.

Lastly, while paper length tends to influence citation counts, especially in the lower quantiles, negatively, the impact of journal impact factor remains consistently positive across all quantiles, with a more significant effect on papers with low to moderate citations. This suggests that for management papers, shorter, more concise articles published in high-impact journals are particularly successful in gaining citations, reflecting the field's preference for accessible, high-impact work that can cater to both academic and practical audiences.

The quantile regression analysis across mathematics, computer science, and management highlights how factors such as author prestige and journal impact factor consistently influence citation impact. Prestige, in particular, emerges as a dominant factor across fields, especially at higher citation levels, where the recognition of well-known authors and prestigious institutions amplifies citation counts. This trend aligns with the established "Matthew Effect" in academia, where visibility and reputation confer advantages that extend beyond the intrinsic quality of the research. While this effect can increase the reach of impactful work, it also underscores an imbalance within academic citation practices. Despite producing rigorous work, authors with lower profiles or those in earlier career stages may struggle to achieve similar citation success due to their limited visibility.

This disparity reflects structural inequalities within academia. The tendency for high-impact journals, established institutions, and recognized authors to receive disproportionately high citations indicates that factors beyond scholarly merit shape the perceived value of research. The Matthew Effect benefits certain researchers and perpetuates an academic environment where influence often stems from reputation rather than research quality. Status biases can further affect peer review and institutional evaluations, frequently favoring known figures and potentially marginalizing innovative contributions from less visible authors. In comparing the Matthew Effect across these three fields, the analysis finds it most assertive in management, emphasizing expertise and reputation in complex, interdisciplinary research contexts.

This background sets the stage for examining the sources and implications of academic inequality, an issue supported by substantial empirical evidence in existing research. The reasons for the existence of the Matthew Effect include the following: (1) the work of renowned scientists is perceived as more significant; (2) their papers are considered more credible; and (3) citing well-known papers and authors can enhance the persuasive power of the citing paper. Peer review is crucial for validating research but may be influenced by "status bias", leading to unequal treatment of papers by well-known versus less-known authors [38]. In economics, there is an institutional Matthew Effect exists whereby the prestige of an author's economics department influences peer recognition of scientific contributions [39]. Repeated papers published in high-impact journals receive, on average, twice as many citations as those published in lower-impact journals. The intrinsic value of a paper is not the sole reason for its citations; the specific Matthew Effect of journals endows papers published in them with additional value beyond their intrinsic quality [40]. Substantial empirical evidence supports the influence of author [41], institution, or journal prestige [42] on citations. Highly cited scientists are also more inclined to collaborate with co-authors who have published multiple highly cited papers [43]. Through quantile regression analysis of papers in three academic fields, we confirmed the universality of the Matthew Effect in mathematics, computer science, and management research. Comparatively, the Matthew Effect, related to prestige factors, is most substantial in management.

4.3. Outcomes of Comparative Analysis

Table 7 presents the importance rankings of factors in predicting papers' citation impact using the XGB, GBDT, RF machine learning methods, and the Linear Regression (LR) model across mathematics, computer science, and management. In the XGB, GBDT,

and RF methods, the rankings of feature importance presented in Table 7 are derived directly from the models based on their calculated importance values. These values represent the contribution of each feature to the model's predictive performance and are normalized for comparability. Features with higher importance values are ranked higher, reflecting their greater influence on the predictions. This ranking approach provides a complementary perspective to the coefficient-based method used in linear regression. The predictive results of the various models are relatively consistent. However, the rankings of a few specific factors exhibit some variation. This suggests that while the primary factors influencing citation impact remain stable across models, certain variables may hold differing levels of importance depending on the analytical approach used.

Table 7. Ranking of key factors influencing citation counts in mathematics, computer science, and management science using XGB, GBDT, RF, and LR methods.

Feature	Math (XGB)	Math (GBDT)	Math (RF)	Math (LR)	CS (XGB)	CS (GBDT)	CS (RF)	CS (LR)	Mgmt (XGB)	Mgmt (GBDT)	Mgmt (RF)	Mgmt (LR)
Prestige	1	1	1	1	1	1	1	1	1	1	1	1
N_auths	8	5	6	3	6	6	4	2	8	4	6	2
N_cntrs	4	7	7	5	4	8	8	6	2	9	8	3
N_catg	2	9	9	8	5	9	9	9	9	8	9	6
N_kwds	6	6	5	6	2	7	7	7	7	7	7	8
N_refs	5	2	2	2	8	3	2	3	6	2	3	4
L_pages	9	3	3	4	7	2	3	4	3	3	2	7
IF	7	8	8	7	3	4	6	5	5	6	5	5
First	3	4	4	9	9	5	5	8	4	5	4	9

In the field of mathematics, regardless of the machine learning method used, prestige consistently emerges as the most crucial factor for citation counts. This finding aligns with results in the fields of management and computer science. This underscores the significance of researchers' prestige and academic prestige in enhancing the citation frequency of a paper. In the GBDT, RF, and LR models, the number of references cited in the paper and article length also exhibit high importance, suggesting that these factors might serve as critical indicators of the depth and breadth of research in mathematics, thereby influencing citation counts. By contrast, the importance of IF and the breadth of categories covered by the paper is relatively low in predicting citation impact in mathematics.

In the field of computer science, prestige also dominates across all models, reflecting the strong influence of computer scientists' prominence on the number of citations a paper receives. The importance of the number of references and article length is also high, with the effect of the number of references being negative. In contrast, the effect of article length is positive. This suggests that in the rapidly evolving field of computer science, the expertise and networks of co-authors significantly impact a paper's influence. The importance of the scope of research topics and keyword usage is lower, indicating that specific research fields and topics are less critical for predicting the citation impact of computer science papers.

In management, the positive effect of prestige and the number of references are significant and notable. In models such as XGB and LR, the importance of the number of collaborating countries increases. This may suggest that in management research, a scholar's prominence, the breadth of their paper collaboration network, and extensive citation of existing literature are vital to achieving high citation counts. This also highlights the complexity of management studies and the need for a broad knowledge base.

The consistent top ranking of prestige across all methods and disciplines underscores its dominant influence on citation impact. However, the variations in rankings for other variables can be attributed to methodological sensitivities, variations in feature interactions, and discipline-specific nuances. For instance, tree-based models emphasize non-linear relationships and feature interactions, while linear regression focuses on mean effects.

Additionally, variables like collaborating countries may gain prominence in management due to their link with international collaboration, whereas paper length may interact with the number of references in computer science to signal technical depth. These differences highlight the importance of considering methodological and disciplinary contexts when interpreting the relative importance of non-content factors.

5. Discussion

5.1. Literature Comparative Analysis

In this study, through a comparative analysis of mathematics, computer science, and management science using multiple methods (XGBoost, Gradient Boosted Decision Trees, Random Forest, and Linear Regression); author prestige consistently emerged as the most influential factor affecting citation counts. Regardless of the discipline or method, author prestige ranked first across all scenarios, highlighting the significant role of academic resource concentration in amplifying visibility and citations. This finding aligns closely with Nielsen and Andersen’s depiction [44] of the stratified distribution of scientific resources and Teixeira da Silva’s emphasis [45] on the “Matthew Effect” in citation practices. For example, consistent with Teixeira da Silva’s assertion [45] that renowned authors amplify their citation impact through greater visibility and academic reputation, this study quantifies the influence of this phenomenon across different fields. Empirically, we also validate Nielsen’s proposition [44] that science operates as a highly stratified social system with highly unequal rewards, offering a quantitative analysis of the substantial citation advantages held by high-prestige authors.

Our findings extend the analysis of citation impact factors conducted by Ahlgren et al. [6]. Beyond discussing factors such as the number of cited references, this study further highlights the role of journal impact factor in shaping citation outcomes, particularly in fields like computer science. Additionally, this work complements Kostea’s ranking [17] of citation impact factors by empirically reinforcing the critical advantage that author prestige offers in securing citations, a finding supported consistently across all disciplines we examined.

While author prestige stands out as the dominant factor in our study, comparisons with other research reveal notable variations in the importance rankings of citation impact factors. To further illustrate these findings, Table 8 summarizes the results of our study alongside rankings and perspectives from other literature.

Table 8. Comparative analysis of citation impact factors.

Factor	Our Study	Rankings and Perspectives in Other Studies
Author Prestige	The most critical factor consistently ranked first, highlighting the core impact of author resources and reputation on citation counts (applicable to all disciplines).	Nielsen and Andersen [44]: The concentration of citation resources among elite scientists causes inequality. Gao et al. [30]: Author h-index and productivity are significant predictors of citations.
Reference Counts	A significant factor across all disciplines, with particularly high impact in mathematics and computer science.	Porwal and Devare [4]: Ranked second, positively correlated with citation counts. Yu et al. [1]: Ranked second in regression coefficients. Gao et al. [30]: Recency also significantly enhances the role of references.
Journal Impact Factor	Ranks high in computer science but has limited impact in mathematics and management science.	Xia et al. [24]: Significant in applied sciences. Li et al. [10]: A crucial predictor of citation counts in biomedical fields. Stegehuis et al. [2]: Second only to early citations in importance.

Table 8. Cont.

Factor	Our Study	Rankings and Perspectives in Other Studies
Early Citations	Moderately important in mathematics, less significant in computer science and management science.	Hu et al. [9]: First two years of citations are the most critical factor. Yu et al. [1]: Early citations are key variables in regression analysis. Stegehuis et al. [2]: The strongest predictor of long-term citation counts.
Abstract Readability	Relatively low importance, with limited impact in certain disciplines.	Porwal and Devare [4]: The most important factor, significantly positively correlated with citation counts.
International Collaboration	Important in management science but less influential in mathematics and computer science.	Gao et al. [30]: Features of international collaboration networks (e.g., PageRank) significantly influence interdisciplinary research. Nielsen and Andersen [44]: Collaboration trends, particularly “super papers”, are increasing.
Network-Related Features	Addressed using author count in this study, highlighting its moderate relevance to citation counts.	Gao et al. [30]: Features such as co-citation network centrality and co-authorship network centrality significantly contribute to citation counts. Zhang et al. [46]: The importance of author network centrality varies by discipline.
Time-Dependent Features	Addressed in some disciplines but of low overall importance.	Hu et al. [9]: Annual citation fluctuations and peak citation-related features are highly important for predicting citation counts. Kosteas [17]: Short-term citations are a key predictor of long-term citations.
Paper Length	Moderately important, with a weaker impact compared to other factors.	Porwal and Devare [4]: Moderately important, positively impacts citation counts.

In summary, our study underscores the central role of author prestige in driving citation counts, while situating this finding within a broader context of diverse citation impact factors. By integrating insights from existing literature, we highlight the multi-faceted nature of citation behaviors shaped by disciplinary norms, network dynamics, and methodological approaches. These findings provide a foundation for more equitable academic evaluation systems that account for these complexities.

5.2. Citation Inequality Stems from Multiple Factors

We attribute citation inequality to three dimensions through an analysis of existing literature: author prestige, reader citation behavior, editorial preferences. First, renowned scholars often possess broader collaboration networks and greater academic influence, granting them higher acceptance rates by editors and more opportunities to publish in high-quality journals, ultimately leading to higher citation rates [45]. In evaluation systems, funding and recruitment decisions heavily rely on bibliometric indicators. Due to their high citation rates and influence, elite scientists are more likely to secure research funding and expand their research labs and collaboration networks, further increasing their publication and citation numbers. This may lead to a greater concentration of funding, publications, and citations [44].

Second, readers’ citation behavior is influenced by social recognition, leading scholars to favor citing highly cited papers even when these papers are not the most optimal or relevant, which contributes to unequal citation distribution [45]. Confirmation bias is a cognitive bias where individuals tend to seek, interpret, or emphasize information that supports their existing beliefs or hypotheses while ignoring or underestimating information that contradicts them. In the scientific community, confirmation bias may manifest as an overvaluation of the work of well-known scientists, perceiving it as more credible or of higher quality, making their work more likely to be cited to enhance the persuasiveness of citing papers. Conversely, the work of emerging or lesser-known scientists may be

subject to excessive scrutiny or outright neglect. Accumulated reputation grants high-profile scholars an advantage in newly published research, creating a positive feedback loop of cumulative advantage [47]. Additionally, as a form of social signal, journal impact factor influences scholars' selection of citation sources, leading to more citations for works published in high-impact journals. The positive feedback loop reinforcing citations for prestigious authors is illustrated in Figure 2.

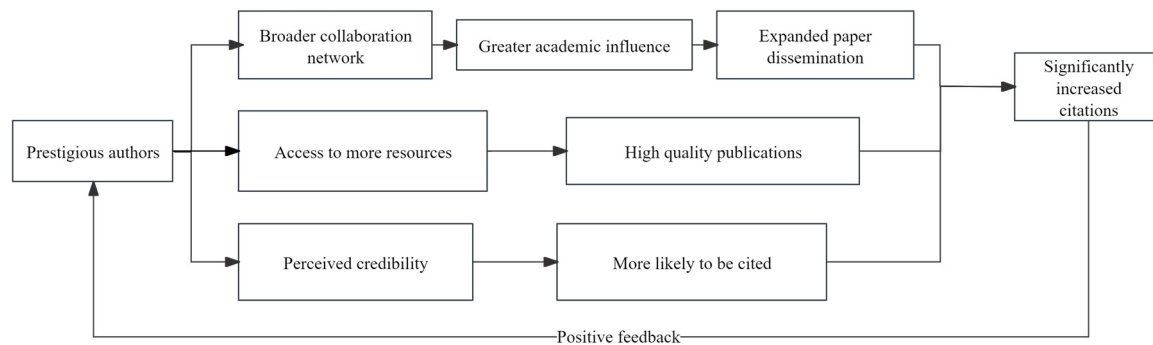


Figure 2. A positive feedback loop reinforcing citations for prestigious authors.

Moreover, journal editors and the peer review process often exhibit a preference for well-known authors, which further exacerbates citation inequality. Editors of high-impact journals are more likely to accept submissions from prestigious institutions or renowned scholars. This preference is not limited to the review process but is also reflected in editorial decisions as the impact factor of journals depends on the citation counts of the articles they publish. Journals often aim to enhance their influence and visibility by publishing works from prominent scholars, thereby reinforcing these authors' visibility and citation advantages [48–50]. This feedback loop is illustrated in Figure 3.

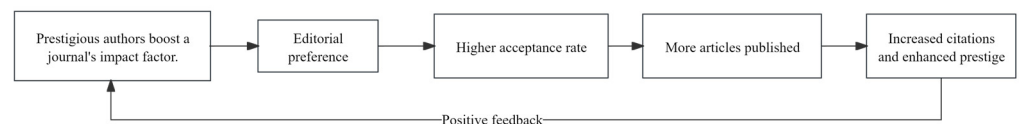


Figure 3. A feedback loop where editors may tend to accept submissions from prestigious authors.

6. Conclusions and Future Directions

6.1. Conclusions

Through a comprehensive comparison of various methods and cross-disciplinary analyses, this study uncovers the underlying mechanisms of citation inequality in academia. The findings reveal that author prestige is a key determinant of citation counts. This conclusion is validated using multiple linear regression, quantile regression, and machine learning techniques such as XGBoost, GBDT, and Random Forest, with consistent results observed across mathematics, computer science, and management. These results address the research question by demonstrating how non-content factors, particularly author prestige, journal impact factors, and collaboration metrics, influence citation impact across disciplines.

Additionally, we delve into the logical pathways underpinning this inequality. High-prestige authors benefit from extensive collaboration networks and strong academic influence, which increase the likelihood of their work being accepted by high-impact journals, thereby enhancing visibility and dissemination. This cumulative advantage creates a positive feedback loop, further consolidating the dominance of prestigious authors within the academic field. Meanwhile, editorial preferences for high-prestige authors exacerbate this inequality, making the academic evaluation system increasingly reliant on identity-

based metrics. Lastly, cross-disciplinary analyses reveal that citation inequality manifests differently across disciplines, influenced by variations in cultural and practical norms. These findings offer valuable insights into the diversity of citation inequality and provide theoretical support for future research.

6.2. Practical Applications

This study offers valuable recommendations for policymakers aiming to foster diversity and inclusivity in academia. Higher education institutions and research funding agencies should reconsider current evaluation standards, reducing the emphasis on identity-based metrics such as author prestige. Instead, these entities should prioritize the quality and innovativeness of research content. Furthermore, introducing multidimensional evaluation frameworks could promote equitable competition and encourage diverse contributions within the academic community.

For academic journal editors, the findings highlight the importance of minimizing favoritism toward high-prestige authors. Editorial boards can enhance transparency and fairness by adopting double-blind peer review processes. Additionally, journals should prioritize publishing innovative, challenging, and exploratory research to push the boundaries of knowledge. These practices would not only mitigate structural inequalities but also create a more inclusive and dynamic academic landscape.

6.3. Future Research Directions

The findings of this study open up opportunities for refining citation prediction models by incorporating additional variables, such as funding sources, collaborative networks, or institutional factors. These variables could complement the structured and textual elements already explored, offering a more comprehensive understanding of the multifaceted drivers of citation behavior. Enhancing model robustness through such expansions would also contribute to a deeper exploration of citation dynamics and further support the development of fairer and more inclusive academic evaluation frameworks.

Additionally, while this study provides valuable insights into structured and textual factors, future research could further explore content-related factors, such as the topics, objectives, or methodologies of cited publications. These aspects are likely to play an important role in shaping citation behavior as researchers often prioritize works that align closely with their thematic interests or provide significant methodological contributions. Integrating these content-related variables could enrich our understanding of the nuanced interplay between content relevance and other determinants of citation impact.

Moreover, textual features such as abstracts, introductions, and titles merit deeper investigation, given their potential influence on how researchers perceive and engage with a publication. Examining dimensions like linguistic style, topic novelty, and research framing could offer richer insights into the mechanisms driving citation behavior. Employing advanced analytical approaches to study the semantic and structural attributes of textual content would further enhance our understanding of how these features contribute to the dissemination and impact of academic work.

Expanding the scope of this research to include a broader range of academic disciplines represents another critical direction for future studies. While this study focuses on mathematics, computer science, and management, disciplines such as the humanities, social sciences, and life sciences may exhibit distinct citation behaviors shaped by their unique norms and practices. Investigating these fields could determine whether citation inequality is a universal phenomenon or context-specific. Interdisciplinary fields may also provide valuable insights into citation dynamics arising from cross-field collaborations, offering a richer perspective on academic impact.

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Abbreviations

The following abbreviations are used in this manuscript:

AMS	American Mathematical Society
ACM	Association for Computing Machinery
IF	Impact factor
LR	Linear regression model
UTD	The University of Texas, Dallas
XGB	Extreme gradient boosting

Appendix A

Table A1. Journals included in this study.

Field	Journals (the Year Founded)
Math	<i>Proceedings of the American Mathematical Society</i> (1950); <i>Transactions of the American Mathematical Society</i> (1900); <i>Contemporary Mathematics</i> (2008); <i>Mathematics of Computation</i> (1943); <i>Memoirs of the American Mathematical Society</i> (1988); <i>St. Petersburg Mathematical Journal</i> (2003); <i>Journal of the American Mathematical Society</i> (1988); <i>Bulletin of the American Mathematical Society</i> (1891); <i>Proceedings of Symposia in Pure Mathematics</i> (1959); <i>CRM Proceedings & Lecture Notes</i> (2000); <i>Fields Institute Communications</i> (2012); <i>Representation Theory</i> (1996); <i>Quarterly of Applied Mathematics</i> (1994); <i>Proceedings of Symposia in Applied Mathematics</i> (1962); <i>Journal of Algebraic Geometry</i> (1992); <i>Electronic Research Announcements of the American Mathematical Society</i> (2008); <i>DIMACS Series in Discrete Mathematics and Theoretical Computer Science</i> (1998); <i>Hyperbolic Problems: Theory, Numerics, and Applications, Part 2</i> (2009); <i>Conformal Geometry and Dynamics</i> (1997); <i>Real and Complex Singularities</i> (2010); <i>Ultrametric Functional Analysis</i> (2003); <i>Topological Algebras and Applications</i> (2013); <i>Arithmetic Geometry, Cryptography, and Coding Theory</i> (1987); <i>Recent Advances in Scientific Computing and Applications</i> (2012); <i>Advances in Non-Archimedean Analysis</i> (2014); <i>Complex Analysis and Dynamical Systems II</i> (2005); <i>Analysis on Graphs and its Applications</i> (2008); <i>Current Trends in Scientific Computing</i> (2003); <i>Advances in Ultrametric Analysis</i> (2018); <i>Function Spaces</i> (1999); <i>Spectral Theory and Mathematical Physics: A Festschrift in Honor of Barry Simon's 60th Birthday</i> (2007); <i>Fractal Geometry and Applications: A Jubilee of Benoit Mandelbrot: Analysis, Number Theory, and Dynamical Systems, Part 1</i> (2004); <i>Noncommutative Geometry and Representation Theory in Mathematical Physics</i> (2005); <i>Complex Analysis and Dynamical Systems III</i> (2008); <i>Harmonic Analysis at Mount Holyoke</i> (2003); <i>High Primes and Misdemeanours: Lectures in Honour of the 60th Birthday of Hugh Cowie Williams</i> (2004); <i>String-Math 2011</i> (2012); <i>Algebraic Structures and their Representations</i> (2013); <i>Asymptotic Methods in Stochastics</i> (2004); <i>Finite Fields: Theory and Applications</i> (1995); <i>Proceedings of Symposia in Pure Mathematics: Algebraic Geometry: Seattle 2005, Volume 80, Parts 1 and 2</i> (2009); <i>Superintegrability in Classical and Quantum Systems</i> (2004); <i>Representations of Algebras and Related Topics</i> (1992); <i>Rings, Modules, and Codes</i> (2019); <i>Gems in Experimental Mathematics</i> (2010); <i>Integrable Systems and Random Matrices in Honor of Percy Deift</i> (2008); <i>Lie Algebras, Vertex Operator Algebras, and their Applications</i> (2007); <i>Operator Theory, Operator Algebras, and Applications</i> (2006); <i>Topics in Finite Fields</i> (2015); <i>Toric Topology</i> (2015); <i>Celebration of the Mathematical Legacy of Raoul Bott</i> (2010); <i>Geometric Analysis of PDE and Several Complex Variables: Dedicated to Francois Trèves</i> (2005); <i>Harmonic Analysis and Partial Differential Equations</i> (1990); <i>Quadratic Forms: Algebra, Arithmetic, and Geometry</i> (2009); <i>Recent Advances in Partial Differential Equations and Applications</i> (2016); <i>Trends in Banach Spaces and Operator Theory</i> (2003); <i>Algebra and its Applications</i> (2000);

Table A1. Cont.

Field	Journals (the Year Founded)
Math	Algebraic and Arithmetic Theory of Quadratic Forms: Proceedings (2004); Banach Algebras and their Applications (2004); Complex Analysis and Potential Theory (2012); Function Spaces in Analysis (2015); Harmonic Maps and Differential Geometry (2011); Probability and Mathematical Physics: A Volume in Honor of Stanislav Molchanov (2007); Quasiconformal Mappings, Riemann Surfaces, and Teichmüller Spaces (2012); Algebraic Methods in Statistics and Probability II (2010); Combinatorial Group Theory, Discrete Groups, and Number Theory (2007); Groups and Model Theory (2012); Groups, Rings, and Group Rings (2009); Interpolation Theory and Applications (2007); Perspectives in Nonlinear Partial Differential Equations: In Honor of Haim Brezis (2007); Recent Advances in Adaptive Computation: Proceedings (2005); Riemann and Klein Surfaces: Automorphisms, Symmetries, and Moduli Spaces (2014); String-Math 2014 (2016); Algebraic Geometry: Salt Lake City 2015, Part 1 (2018); Anatomy of Integers (2008); Complex Analysis and Dynamical Systems VII (2017); Function Spaces in Modern Analysis (2011); Geometry and Topology of Manifolds (2005); Geometry, Groups, and Dynamics (2015); Graphs and Discovery (2005); Groups and Symmetries: From Neolithic Scots to John McKay (2009); Number Theory (1969); Recent Advances in Differential Equations and Mathematical Physics (2006); Rings, Modules, and Representations (2009); Stable and Unstable Homotopy (1998); Topology of Algebraic Varieties and Singularities (2011); Algebras, Representations, and Applications (2009); Categories in Algebra, Geometry, and Mathematical Physics (2007); Combinatorics and Graphs (2010); Commutative Algebra: Interactions with Algebraic Geometry (2003); Complex Analysis and Dynamical Systems IV, Part 1: Function Theory and Optimization (2011); Complex Analysis and Dynamical Systems V (2013); Complex Analysis and Dynamical Systems VI, Part 2: Complex Analysis, Quasiconformal Mappings, Complex Dynamics (2016); Error-Correcting Codes, Finite Geometries, and Cryptography (2010); Finite Fields and Applications (2008); Geometry of Submanifolds (2020); Groups, Rings, and Algebras (2006); Idempotent Mathematics and Mathematical Physics (2005); In the Tradition of Ahlfors and Bers III (2004); Jack, Hall-Littlewood, and Macdonald Polynomials (2006); Mathematical Studies on Human Disease Dynamics: Emerging Paradigms and Challenges (2006); Quantum Graphs and their Applications (2006); Ring Theory and its Applications (2014); Spectral Analysis in Geometry and Number Theory (2009); String-Math 2013 (2014); Algebraic Geometry: Salt Lake City 2015, Part 2 (2018); Geometric Methods in Group Theory (2005); Geometry of Riemann Surfaces and Abelian Varieties (2006); Group Theory and Numerical Analysis (2005); Groups, Algebras, and Applications (2011); Mathematical Problems in Quantum Physics (2018); Panorama of Mathematics: Pure and Applied (2016); Pseudo-Differential Operators, Partial Differential Equations, and Time-Frequency Analysis (2007); Recent Trends in Nonlinear Partial Differential Equations II: Stationary Problems (2013); Surveys on Discrete and Computational Geometry: Twenty Years Later (2008); Tropical and Idempotent Mathematics (2009); Advances in Switching Networks (1998); Affine Algebraic Geometry: The Russell Festschrift (2011); Algebraic and Topological Dynamics (2005); Complex Analysis and Dynamical Systems (2004); Complex Analysis and Dynamical Systems VI, Part 1: PDE, Differential Geometry, Radon Transform (2015); Diophantine Methods, Lattices, and Arithmetic Theory of Quadratic Forms (2013); Group Representations, Ergodic Theory, and Mathematical Physics: A Tribute to George W. Mackey (2008); Hyperbolic Problems: Theory, Numerics, and Applications, Part 1 (2009); Interactions of Classical and Numerical Algebraic Geometry (2009); Models, Logics, and Higher-Dimensional Categories: A Tribute to the Work of Mihály Makkai (2011); Noncommutative Rings and their Applications (2015); Ordered Algebraic Structures and Related Topics (2017); Unusual Applications of Number Theory (2004); Additive Combinatorics (2007); Adventures in Mathematical Physics (2007); Algebra, Geometry, and their Interactions (2007); Automorphic Forms and Related Geometry: Assessing the Legacy of I.I. Piatetski-Shapiro (2014); Complex Analysis and Dynamical Systems IV, Part 2: General Relativity, Geometry, and PDE (2011); Groups, Rings, Group Rings, and Hopf Algebras (2017); In the Tradition of Ahlfors and Bers IV (2007); In the Tradition of Ahlfors and Bers V (2010); Modern Trends in Constructive Function Theory (2016); Operator Algebras, Quantization, and Noncommutative Geometry (2004); Recent Trends in Nonlinear Partial Differential Equations I: Evolution Problems (2013); Representation Theory BK (2009); String-Math 2012 (2015); Tapas in Experimental Mathematics (2008); Topological and Asymptotic Aspects of Group Theory (2006); Advances in p-Adic and Non-Archimedean Analysis (2010); Advances in the Theory of Automorphic Forms and their L-Functions (2016); Affine Algebraic Geometry (2005); Communicating Mathematics (2009); Finite Simple Groups: Thirty Years of the Atlas and Beyond (2017); Fluids and Waves: Recent Trends in Applied Analysis (2007); Frames and Harmonic Analysis (2018); Geometric and Probabilistic Structures in Dynamics (2008); Geometry and Topology Down Under (2013); Ischia Group Theory 2004: Proceedings (2006); Lie Algebras and Related Topics (1990); New Trends in Noncommutative Algebra (2012); Operator Structures and Dynamical Systems (2009); Optimization: Theory and Related Topics (2012); Representation Theory and Mathematical Physics (2011); Theory and Applications of Finite Fields (2012); Topological Recursion and its Influence in Analysis, Geometry, and Topology (2018); Ubiquitous Heat Kernel (2006); Categorical, Homological, and Combinatorial Methods in Algebra (2020); Character Theory of Finite Groups (1976); Control Methods in PDE Dynamical Systems (2007); Graph Partitioning and Graph Clustering (2013); Hilbert Spaces of Analytic Functions (2010); Homotopy Theory of Function Spaces and Related Topics (2010); In the Tradition of Ahlfors Bers VI (2013); Infinite Products of Operators and Their Applications (2015); Local and Global Methods in Algebraic Geometry (2018); Logic and Its Applications (2005); Mathematics of Continuous and Discrete Dynamical Systems (2014); Motives and Algebraic Cycles: A Celebration in Honour of Spencer J. Bloch (2009); Multiple Dirichlet Series, Automorphic Forms and Analytic Number Theory (2006); Noncommutative Geometry and Global Analysis (2011); Nonlinear Analysis and Optimization I: Nonlinear Analysis (2010);

Table A1. Cont.

Field	Journals (the Year Founded)
Math	<i>Nonlinear Analysis and Optimization II: Optimization</i> (2010); <i>Poisson Geometry in Mathematics and Physics</i> (2008); <i>Recent Progress on Some Problems in Several Complex Variables and Partial Differential Equations</i> (2006); <i>Recent Trends in Ergodic Theory and Dynamical Systems</i> (2015); <i>Regulators</i> (2012); <i>Scholar: A Scientific Celebration Highlighting Open Lines of Arithmetic Research</i> (2015); <i>Singularities in PDE and the Calculus of Variations</i> (2008); <i>Spectral Geometry</i> (2012); <i>Symmetry in Mathematics and Physics</i> (2009); <i>Topics in Functional Analysis and Algebra</i> (2016); <i>Tropical and Idempotent Mathematics and Applications</i> (2014); <i>Universality and Renormalization: From Stochastic Evolution to Renormalization of Quantum Fields</i> (2007); <i>Vertex Operator Algebras and Related Areas</i> (2009); <i>WIN: Women in Numbers, Research Directions in Number Theory</i> (2011); <i>Alpine Expedition Through Algebraic Topology</i> (2014); <i>Analysis, Geometry, and Quantum Field Theory</i> (2012); <i>Commutative Algebra and Algebraic Geometry</i> (2005); <i>Commutative Algebra and Its Connections to Geometry</i> (2011); <i>Data Depth: Robust Multivariate Analysis, Computational Geometry, and Applications</i> (2006); <i>Dynamical Systems, Ergodic Theory, and Probability in Memory of Kolya Chernov</i> (2017); <i>Dynamics, Topology, and Numbers</i> (2020)
Computer Science	<i>ACM SIGPLAN Notices</i> (1971); <i>ACM Transactions on Graphics</i> (1982); <i>Communications of the ACM</i> (1958); <i>Proceedings of the VLDB Endowment</i> (2008); <i>ACM Computing Surveys</i> (1969); <i>ACM SIGCOMM Computer Communication Review</i> (1995); <i>ACM Transactions on Multimedia Computing Communications and Applications</i> (2005); <i>ACM Transactions on Embedded Computing Systems</i> (2002); <i>ACM Transactions on Design Automation of Electronic Systems</i> (1996); <i>Proceedings of the ACM on Programming Languages</i> PACMPL (2017); <i>ACM Transactions on Intelligent Systems and Technology</i> (2010); <i>ACM Transactions on Knowledge Discovery from Data</i> (2007); <i>Journal of the ACM</i> (1954); <i>ACM Transactions on Sensor Networks</i> (2005); <i>ACM Transactions on Architecture and Code Optimization</i> (2004); <i>ACM Transactions on Information Systems</i> (1983); <i>Proceedings of the ACM on Interactive Mobile Wearable and Ubiquitous Technologies</i> IMWUT (2017); <i>SIGMOD Record</i> (1969); <i>ACM Transactions on Algorithms</i> (2005); <i>ACM Transactions on Software Engineering and Methodology</i> (1992); <i>ACM Transactions on Mathematical Software</i> (1975); <i>ACM Transactions on Asian and Low Resource Language Information Processing</i> (2015); <i>ACM Transactions on Internet Technology</i> (2001); <i>ACM Transactions on Computer Human Interaction</i> (1994); <i>ACM Journal on Emerging Technologies in Computing Systems</i> (2005); <i>ACM Transactions on Computational Logic</i> (2000); <i>ACM Transactions on Database Systems</i> (1976); <i>ACM Transactions on Programming Languages and Systems</i> (1979); <i>ACM Transactions on Reconfigurable Technology and Systems</i> (2008); <i>ACM Journal on Computing and Cultural Heritage</i> (2008); <i>ACM Transactions on Modeling and Computer Simulation</i> (1991); <i>ACM Transactions on the Web</i> (2007); <i>ACM Transactions on Applied Perception</i> (2004); <i>ACM Transactions on Storage</i> (2005); <i>ACM Transactions on Autonomous and Adaptive Systems</i> (2006); <i>ACM Transactions on Computing Education</i> (2010); <i>Data Base for Advances in Information Systems</i> (1969); <i>Proceedings of the ACM on Measurement and Analysis of Computing Systems</i> (2021); <i>ACM Transactions on Computer Systems</i> (1983); <i>ACM Transactions on Privacy and Security</i> (2016); <i>Proceedings of the ACM on Computer Graphics and Interactive Techniques</i> (2018); <i>ACM Transactions on Interactive Intelligent Systems</i> (2011); <i>ACM Transactions on Cyber Physical Systems</i> (2017); <i>ACM Transactions on Human Robot Interaction</i> (2018); <i>ACM Transactions on Information and System Security</i> (1999); <i>ACM Transactions on Management Information Systems</i> (2010); <i>ACM Transactions on Spatial Algorithms and Systems</i> (2015); <i>ACM Journal of Data and Information Quality</i> (2009); <i>ACM Transactions on Internet of Things</i> (2020); <i>Computer Graphics US</i> (1978); <i>ACM Transactions on Parallel Computing</i> (2014); <i>ACM Transactions on Accessible Computing</i> (2008); <i>ACM Transactions on Computation Theory</i> (2009); <i>GetMobile Mobile Computing Communications Review</i> (1997); <i>ACM Communications in Computer Algebra</i> (2011); <i>ACM Transactions on Economics and Computation</i> (2013); <i>Formal Aspects of Computing</i> (1989); <i>ACM Transactions on Quantum Computing</i> (2020); <i>ACM Transactions on Modeling and Performance Evaluation of Computing Systems</i> (2016); <i>Applied Computing Review</i> (1993); <i>ACM SIGECOM Exchanges</i> (2000)
Mgmt	<i>Accounting Review</i> (1926); <i>Journal of Accounting and Economics</i> (1979); <i>Journal of Accounting Research</i> (1963); <i>Journal of Finance</i> (1946); <i>Journal of Financial Economics</i> (1974); <i>Review of Financial Studies</i> (1986); <i>Information Systems Research</i> (1990); <i>INFORMS Journal on Computing</i> (1989); <i>MIS Quarterly</i> (1977); <i>Journal of International Business Studies</i> (1970); <i>Journal of Consumer Research</i> (1974); <i>Journal of Marketing</i> (1936); <i>Journal of Marketing Research</i> (1964); <i>Marketing Science</i> (1971); <i>Academy of Management Journal</i> (1958); <i>Academy of Management Review</i> (1978); <i>Administrative Science Quarterly</i> (1956); <i>Management Science</i> (1954); <i>Strategic Management Journal</i> (1980); <i>Journal of Operations Management</i> (1980); <i>Manufacturing & Service Operations Management</i> (1999); <i>Operations Research</i> (1952); <i>Production and Operations Management</i> (1992); <i>Organization Science</i> (1990)

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